

MQTT Broker Complete Guide

localhost (Aedes) vs HiveMQ

Project: IoT Project 2

Hardware: ESP8266 NodeMCU + Node-RED

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Date: March 2026

1. MQTT Kya Hai?

MQTT (Message Queuing Telemetry Transport) ek lightweight messaging protocol hai jo IoT devices ke liye design kiya gaya hai. Yeh Publish/Subscribe model pe kaam karta hai jisme devices data bhejte aur receive karte hain ek central 'Broker' ke through.

1.1 MQTT Architecture

Component	Role	Hamare Project Mein
Publisher	Data bhejta hai	ESP8266 NodeMCU
Broker	Messages route karta hai	Aedes (local) / HiveMQ
Subscriber	Data receive karta hai	Node-RED Dashboard
Topic	Channel/address	tempNode, humiNode etc.

1.2 Hamare Project ke MQTT Topics

Topic Name	Direction	Data
tempNode	ESP → Node-RED	Temperature °C (e.g. 30.9)
humiNode	ESP → Node-RED	Humidity % (e.g. 26)
ldrNode	ESP → Node-RED	Light / Dark
mq135Node	ESP → Node-RED	Air Quality 0-1024 (ppm)
distanceNode	ESP → Node-RED	Distance in cm
led1StatusNode	ESP → Node-RED	ON / OFF
led2ControlNode	Node-RED → ESP	1/ON or 0/OFF

2. localhost — Aedes MQTT Broker

Aedes Broker Kya Hai?

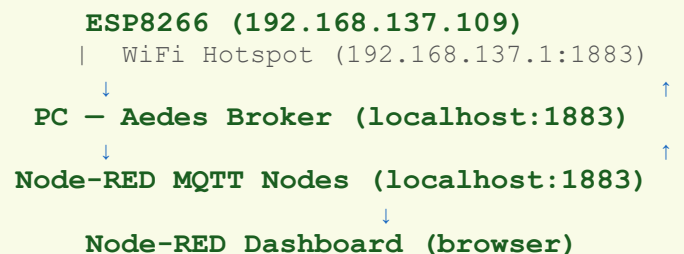
Aedes ek Node-RED plugin hai jo directly Node-RED ke andar ek MQTT broker run karta hai. Iska matlab hai ki alag se koi software install karne ki zaroorat nahi — Node-RED khud hi broker ban jata hai!

2.1 Aedes Broker Kaise Kaam Karta Hai

Jab aap Node-RED mein Aedes broker node add karte hain, tab:

- Node-RED ek MQTT server start karta hai port 1883 pe
- Yeh server sirf local machine pe available hota hai
- ESP8266 PC ke hotspot IP se connect hota hai (e.g. 192.168.137.1)
- Node-RED MQTT nodes localhost se connect hote hain
- Dono same Aedes broker se baat karte hain

2.2 Network Diagram — Aedes (localhost)



2.3 Aedes Setup — Step by Step

Step 1: Plugin Install karo

Node-RED mein jao:

```

Menu (≡) → Manage Palette → Install Tab
Search: node-red-contrib-aedes
Install button click karo → Done

```

Step 2: Aedes Broker Node Add karo

- Left panel mein 'aedes broker' dhundho
- Canvas pe drag karo
- Double click karo
- Name: localhost, MQTT Port: 1883
- Done → Deploy

Step 3: MQTT IN Nodes Configure karo

Har MQTT IN node mein (Temperature, Humidity etc.):

- Server: 192.168.137.1 (PC Hotspot IP)
- Port: 1883
- Ya '+' button se naya broker add karo

Step 4: ESP8266 Code

```
const char* mqtt_server = "192.168.137.1"; // PC Hotspot IP
const int mqtt_port = 1883;
```

2.4 Aedes ke Fayde aur Nuksan

☑ FAYDE (Advantages)	✗ NUKSAN (Disadvantages)
Internet ki zaroorat nahi	Sirf local network mein kaam karta hai
Bahut fast — low latency	PC band ho to kaam nahi karta
Free — koi subscription nahi	Bahar se access nahi ho sakta
Alag software install nahi	Node-RED hamesha ON rakhna padta hai
Data private — bahar nahi jata	Multiple PC/location pe kaam nahi
Offline projects ke liye perfect	Production use ke liye suitable nahi

2.5 Aedes Troubleshooting

Problem	Solution
Nodes 'connecting' dikha rahe hain	MQTT node mein Server ko 192.168.137.1 karo (localhost nahi)
ESP connect nahi ho raha	Hotspot IP check karo — CMD mein 'ipconfig' chalao
Dashboard empty hai	Deploy dobara karo, topics match karo
Aedes node nahi dikh raha	node-red-contrib-aedes install karo Manage Palette se

3. HiveMQ — Cloud MQTT Broker

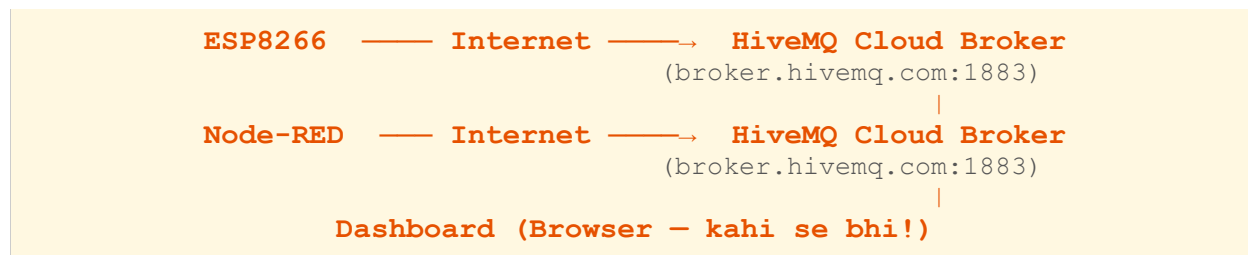
HiveMQ Kya Hai?

HiveMQ ek public cloud MQTT broker hai jo internet pe available hai. Koi bhi free mein use kar sakta hai — koi login ki zaroorat nahi. Yeh globally accessible hai, matlab ESP8266 kisi bhi network se connect ho sakta hai.

3.1 HiveMQ Details

Property	Value
Broker Address	broker.hivemq.com
MQTT Port	1883 (non-secure)
MQTT Port (SSL)	8883 (secure)
WebSocket Port	8000
Login Required	Nahi — completely anonymous
Cost	Free (public broker)
Message Limit	Unlimited (fair use)
Availability	24/7 globally available

3.2 Network Diagram — HiveMQ (Cloud)



3.3 HiveMQ Setup — Step by Step

ESP8266 Arduino Code

```
//const char* mqtt_server = "192.168.137.1"; // Local — comment out karo
const char* mqtt_server = "broker.hivemq.com"; // HiveMQ
const int mqtt_port = 1883;
```

Node-RED MQTT Nodes Configure karo

- Har MQTT node pe double click karo
- Server dropdown mein 'HiveMQ' select karo

- Ya naya broker add karo: broker.hivemq.com, Port: 1883
- Deploy karo

3.4 HiveMQ ke Fayde aur Nuksan

☑ FAYDE (Advantages)	✗ NUKSAN (Disadvantages)
Internet se kahi bhi access ho sakta	Internet connection must hai
Koi setup nahi — seedha use karo	Public broker — data secure nahi
Multiple devices worldwide connect ho sakte	Latency thodi zyada (internet delay)
PC hotspot ki zaroorat nahi	Koi bhi same topic subscribe kar sakta
Demo aur testing ke liye perfect	Server down ho to project band
Free mein cloud MQTT milta hai	Offline use nahi ho sakta

4. Localhost (Aedes) vs HiveMQ — Full Comparison

Feature	Aedes (localhost)	HiveMQ (Cloud)
Internet Required?	✗ Nahi	✓ Haan
Speed/Latency	🚀 Bahut Fast (local)	⚡ Medium (internet)
Setup Complexity	Medium (plugin install)	Easy (seedha use)
Remote Access	✗ Sirf local	✓ Worldwide
Data Privacy	✓ 100% Private	✗ Public broker
Reliability	PC pe depend karta hai	HiveMQ servers pe depend
Cost	Free	Free (public)
Multiple Networks	✗ Ek network only	✓ Any network
Best For	Home/Lab projects	Demo/Remote projects
Hamare Project Mein	✓ Currently USE kar rahe	Pehle use kiya tha

4.1 Kab Kya Use Karein?

Aedes (localhost) use karo jab:

- Ghar ya lab mein kaam kar rahe ho
- Internet nahi ho ya slow ho
- Data private rakhna ho
- Fast response time chahiye
- Offline demo deni ho

HiveMQ use karo jab:

- Kahin bhi remote access chahiye
- Multiple locations pe devices hain
- Quick demo ya testing karni ho
- Internet hamesha available ho
- Simple setup chahiye bina kisi plugin ke

5. Code Mein Switch Karna — Aedes ↔ HiveMQ

5.1 Arduino Code Switch

Sirf ek line change karni hai:

HiveMQ ke liye:

```
const char* mqtt_server = "broker.hivemq.com";  
// Koi WiFi hotspot zaroorat nahi  
// ESP8266 kisi bhi WiFi se connect ho sakta
```

Aedes (localhost) ke liye:

```
const char* mqtt_server = "192.168.137.1"; // PC Hotspot IP  
// ESP8266 ko RSMaurya hotspot se connect karo  
// PC pe Node-RED + Aedes chal raha hona chahiye
```

5.2 Node-RED Switch

Har MQTT node mein Server change karo:

Setting	Aedes (local)	HiveMQ (cloud)
Server	192.168.137.1	broker.hivemq.com
Port	1883	1883
Username	(blank)	(blank)
Password	(blank)	(blank)

6. Hamare Project ki Current Status

Currently Chal Raha Hai:

Aedes localhost broker chal raha hai aur ESP8266 se live data aa raha hai dashboard pe!

6.1 Working Configuration

Parameter	Value
WiFi SSID	RSMaurya
ESP8266 IP	192.168.137.109
PC (Broker) IP	192.168.137.1
MQTT Broker	Aedes (Node-RED Plugin)
MQTT Port	1883
Dashboard URL	http://127.0.0.1:1880/ui
Node-RED URL	http://127.0.0.1:1880

6.2 Live Sensor Data (Tested Values)

Sensor	Value	Unit	Status
Temperature	31.1	°C	☒ Working
Humidity	27	%	☒ Working
Air Quality	80	ppm	☒ Working
Distance	16.08	cm	☒ Working
LDR	Light	text	☒ Working
LED2 Control	ON/OFF	switch	☒ Working

7. Author Information

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